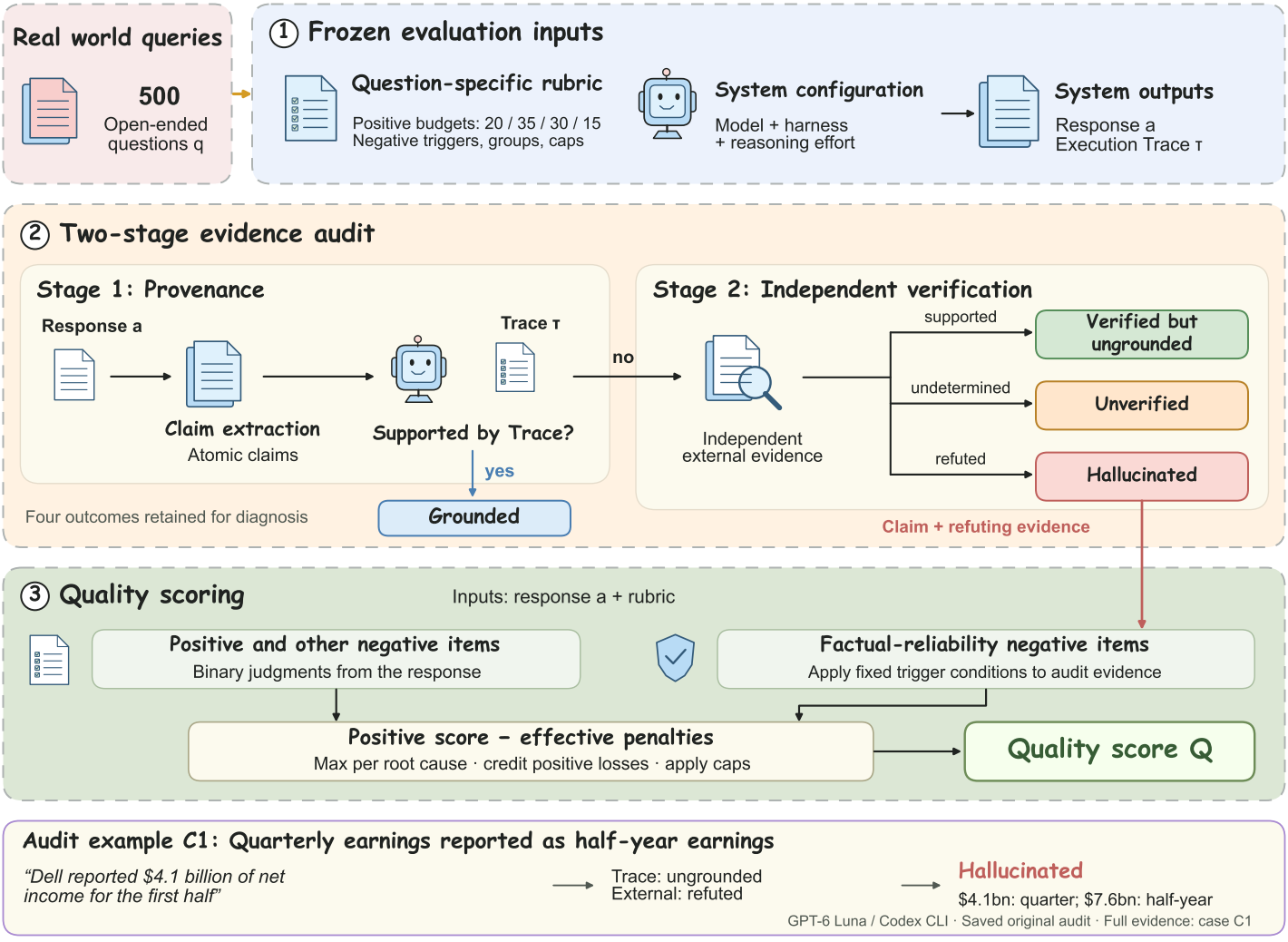

Real world queries
500
Open-ended questions q
① Frozen evaluation inputs
Question-specific rubric
Positive budgets: 20 / 35 / 30 / 15
Negative triggers, groups, caps
System configuration
Model + harness
+ reasoning effort
System outputs
Response a
Execution Trace τ
② Two-stage evidence audit
Stage 1: Provenance
Response a
Claim extraction
Atomic claims
Trace τ
Supported by Trace?
yes
Grounded
Four outcomes retained for diagnosis
no
Stage 2: Independent verification
Independent external evidence
supported
Verified but ungrounded
undetermined
Unverified
refuted
Hallucinated
Claim + refuting evidence
③ Quality scoring
Inputs: response a + rubric
Positive and other negative items
Binary judgments from the response
Factual-reliability negative items
Apply fixed trigger conditions to audit evidence
Positive score − effective penalties
Max per root cause · credit positive losses · apply caps
Quality score Q
Audit example C1: Quarterly earnings reported as half-year earnings
"Dell reported $4.1 billion of net income for the first half"
Trace: ungrounded
External: refuted
Hallucinated
$4.1bn: quarter; $7.6bn: half-year
GPT-6 Luna / Codex CLI · Saved original audit · Full evidence: case C1